

PRACTICAL 7

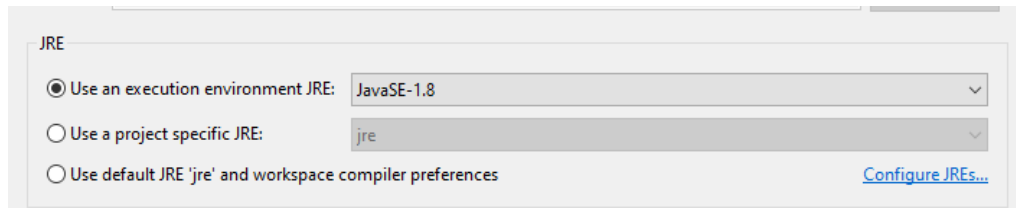
Write and test a program to provide a total number of objects present / available on the page.

Step 1: Download Selenium Server.

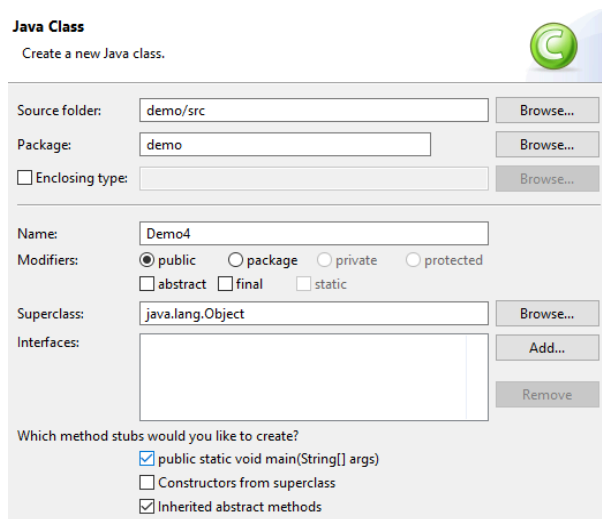
Step 2: Download the geckodriver.

Step 3: Download Selenium Standalone jar.

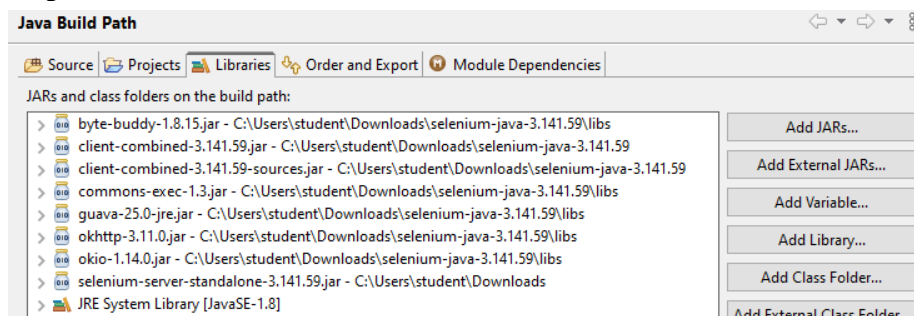
Step 4: Create a java project.



Step 5: Create a package → Click on Public static main Checkbox → Create class.



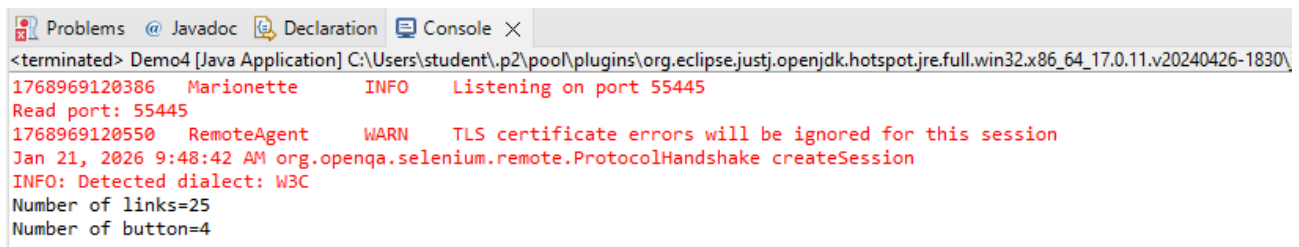
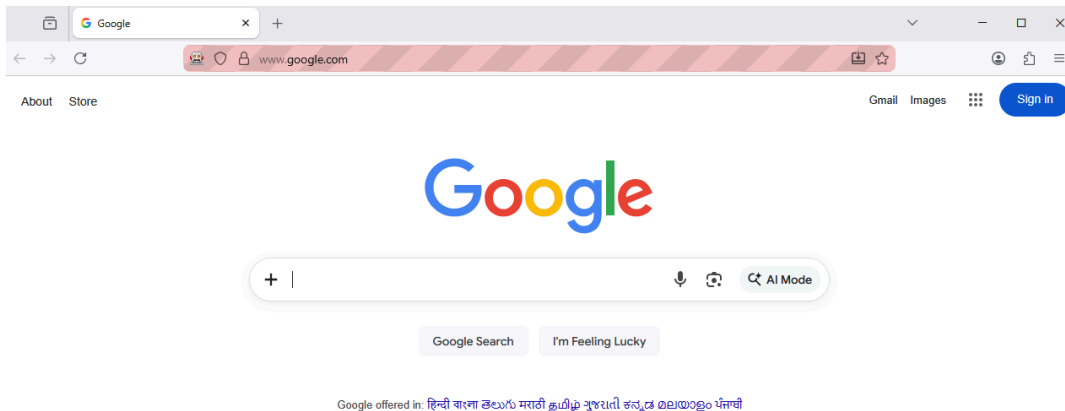
Step 6: Add external libraries.



Step 7: Write code

```
package demo;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.By;
import org.openqa.selenium.WebElement;
import java.util.List;
public class Demo4 {
    static String driverPath =
        "C:\\Users\\student\\Downloads\\geckodriver-v0.34.0-win32\\geckodriver.exe";
    public static void main(String[] args) {
        System.setProperty("webdriver.gecko.driver", driverPath);
        FirefoxOptions options = new FirefoxOptions();
        WebDriver driver = new FirefoxDriver(options);
        driver.get("https://www.google.com/");
        List<WebElement> mylist = driver.findElements(By.xpath("//a"));
        List<WebElement> mylist1 = driver.findElements(By.xpath("//input[@type='submit']"));
        System.out.println("Number of links=" + mylist.size());
        System.out.println("Number of button=" + mylist1.size());
    }
}
```

Output:



PRACTICAL 8

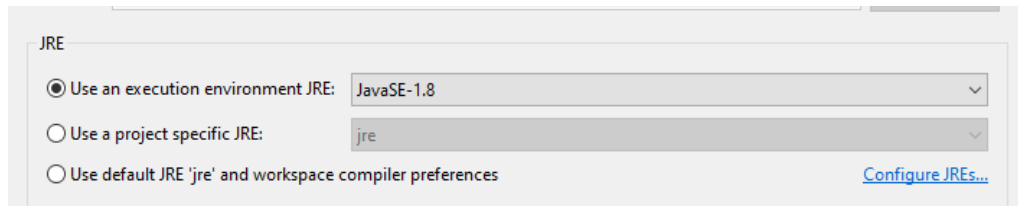
Write and test a program to get the number of items in a list / combo box.

Step 1: Download Selenium Server.

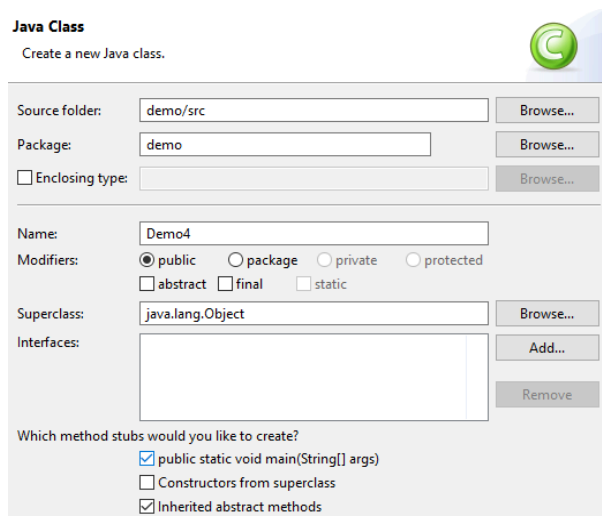
Step 2: Download the geckodriver.

Step 3: Download Selenium Standalone jar.

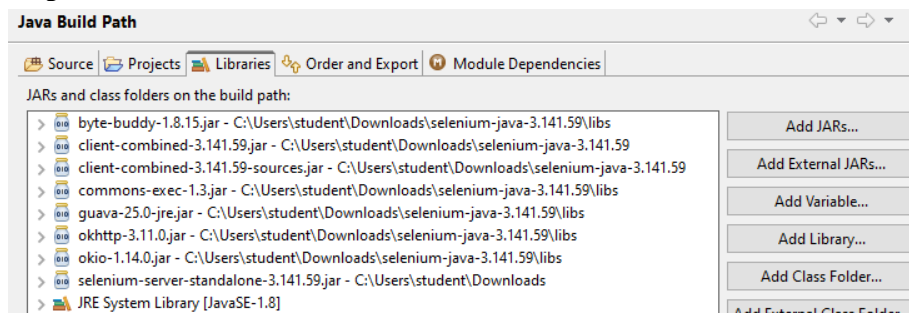
Step 4: Create a java project.



Step 5: Create a package → Click on Public static main Checkbox → Create class.



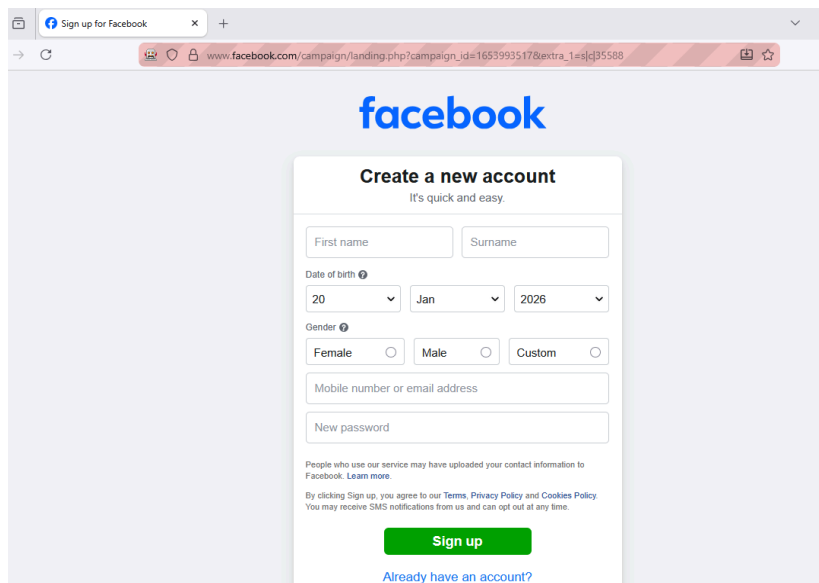
Step 6: Add external libraries.



Step 7: Write code

```
package demo;
import java.util.*;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.support.ui.Select;
public class Demo4 {
    public static void main(String[] args) {
        System.setProperty("webdriver.gecko.driver","C:\\Users\\student\\Downloads\\geckodriver-v0.34.0-win32\\geckodriver.exe");
        WebDriver driver = new FirefoxDriver();
        driver.get("https://www.facebook.com/campaign/landing.php?campaign_id=1653993517&extra_1=s|c|35588");
        Select se = new Select(driver.findElement(By.xpath("//select[@id='month']")));
        List<WebElement> mylist = se.getOptions();
        mylist.size();
        System.out.println("Number of items = "+mylist.size());
    }
}
```

Output:



```
Problems @ Javadoc Declaration Console X
<terminated> Demo4 [Java Application] C:\Users\student\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_17.0.11.v2024
1768969840372 marionette INFO Listening on port 55601
Read port: 55601
1768969840469 RemoteAgent WARN TLS certificate errors will be ignored for this session
Jan 21, 2026 10:00:42 AM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
JavaScript warning: https://static.xx.fbcdn.net/rsrsc.php/v4ihVQ4/yW/1/en_GB-j/vw1h9pbSv6z.js, line 36: WEBGL
console.warn: LoginRecipes: "Falling back to a synchronous message for: https://www.facebook.com."
Number of items = 12
```

Using TestNG

Step 1: Download Selenium Server.

Step 2: Download the geckodriver.

Step 3: Download Selenium Standalone jar.

Step 4: Create a java project.

Step 5: Install TestNG. and Run as TestNG.

Code:

```
package demo11;
import java.util.List;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.support.ui.Select;
import org.testng.annotations.AfterClass;
import org.testng.annotations.BeforeClass;
import org.testng.annotations.Test;
public class combo1 {
    WebDriver driver;
    @BeforeClass
    public void setUp() {
        System.setProperty(
            "webdriver.gecko.driver",
            "C:\\Users\\student\\Downloads\\geckodriver-v0.34.0-win32\\geckodriver.exe");
        driver = new FirefoxDriver();
        driver.get(
            "https://www.facebook.com/campaign/landing.php?campaign_id=1653993517&extra_1=s|c|35588");
    }
    @Test
    public void countMonthDropdownItems() {
        Select se = new Select(driver.findElement(By.id("month")));
        List<WebElement> mylist = se.getOptions();
        System.out.println("Number of items = " + mylist.size());
    }
    @AfterClass
    public void tearDown() {
        driver.quit();
    }
}
```

Output:

```
Problems @ Javadoc Declaration Console × Progress Results of running class demo5
<terminated> demo5 [TestNG] C:\Users\student\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_
Number of items = 12
PASSED: countMonthDropdownItems

=====
Default test
Tests run: 1, Failures: 0, Skips: 0
=====

=====
Default suite
Total tests run: 1, Failures: 0, Skips: 0
=====
```

PRACTICAL 9

Write and test a program to count the number of check boxes on the page checked and unchecked count.

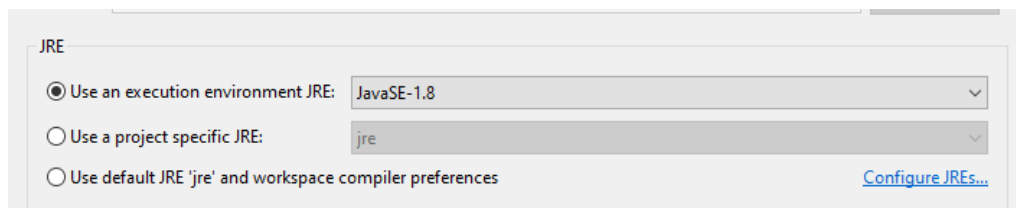
Step 1: Create your own registration form(with some checked and unchecked checkbox).

Step 2: Download Selenium Server.

Step 3: Download the geckodriver.

Step 4: Download Selenium Standalone jar.

Step 5: Create a java project.



Step 6: Use default package → Create class.

Step 7: Add external libraries.

Step 8: Write Code.

register.html

```
<!DOCTYPE html>
<html>
<body>
<form>
<h2>Text Input</h2>First name:<br>
<input type="text" name="firstname"><br>Last name:<br>
<input type="text" name="lastname"><br>
<h2>Select Gender</h2>
<input type="radio" name="gender" value="male" checked> Male<br>
<input type="radio" name="gender" value="female"> Female<br>
<input type="radio" name="gender" value="other"> Other<br>
<h2>Select Languages Known</h2>
<input type="checkbox" name="lang" value="Java"> Java<br>
<input type="checkbox" name="lang" value="PHP" checked="checked"> PHP<br>
<input type="checkbox" name="lang" value="C#"> C#<br>
<input type="checkbox" name="lang" value="Python" checked="checked"> Python<br>
<input type="checkbox" name="lang" value="Ruby"> Ruby<br><br><br>
<input type="submit" value="Submit">
</form>
</body>
</html>
```

demo5.java

```
import java.util.*;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
class demo5
{
    public static void main(String args[])
    {
        System.setProperty("webdriver.gecko.driver","C:\\Users\\student\\Downloads\\geckodriver-v0.34.0-win32\\geckodriver.exe");
        WebDriver driver = new FirefoxDriver();
        driver.get("file:///C:/Users/New%20folder/CS(TY)/Prac/prac%209/register.html");
        int chk = 0;
        int unchk = 0;
        List <WebElement> els = driver.findElements(By.xpath("//input[@type='checkbox']"));
        for(WebElement el : els)
        {
            if(el.isSelected()) {
                chk++;
            }
            else{
                unchk++;
            }
        }
        System.out.println("Total checked items" + chk);
        System.out.println("Total unchecked items" + unchk);
    }
}
```

Output:



Text Input

First name:

Last name:

Select Gender

- ☒ Male
☐ Female
☐ Other

Select Languages Known

- ☐ Java
☒ PHP
☐ C#
☒ Python
☐ Ruby

```
Total checked items2  
Total unchecked items3
```

Using TestNG

Step 1: Download Selenium Server.

Step 2: Download the geckodriver.

Step 3: Download Selenium Standalone jar.

Step 4: Create a java project.

Step 5: Use default package → Create Class.

Step 5: Install TestNG. and Run as TestNG.

Code:

```
import java.util.List;  
import org.openqa.selenium.By;  
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.WebElement;  
import org.openqa.selenium.firefox.FirefoxDriver;  
import org.testng.annotations.AfterTest;  
import org.testng.annotations.BeforeTest;  
import org.testng.annotations.Test;  
public class checkboxTestNG {  
    WebDriver driver;  
    @BeforeTest  
    public void setup() {  
        System.setProperty("webdriver.gecko.driver",  
            "C:\\\\Users\\student\\Downloads\\geckodriver-v0.34.0-win32\\geckodriver.exe");  
        driver = new FirefoxDriver();  
        driver.get("https://the-internet.herokuapp.com/checkboxes");  
    }  
    @Test  
    public void countCheckedAndUncheckedBoxes() {  
        int checked = 0;  
        int unchecked = 0;  
        List<WebElement> checkboxes =  
            driver.findElements(By.xpath("//input[@type='checkbox']"));  
        for (WebElement checkbox : checkboxes) {  
            if (checkbox.isSelected()) {  
                checked++;  
            } else {  
                unchecked++;  
            }  
        }  
        System.out.println("Total checked items: " + checked);  
    }  
}
```

```
        System.out.println("Total unchecked items: " + unchecked);
    }

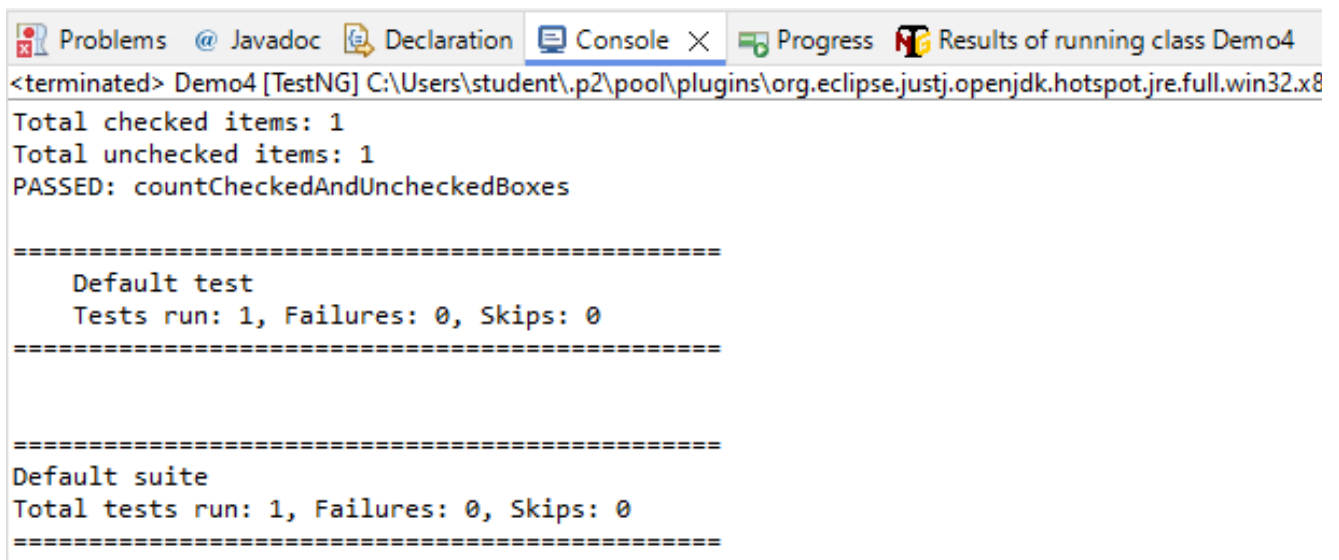
    @AfterTest
    public void tearDown() {
        driver.quit();
    }
}
```

Output:



Checkboxes

- ☐ checkbox 1
☒ checkbox 2



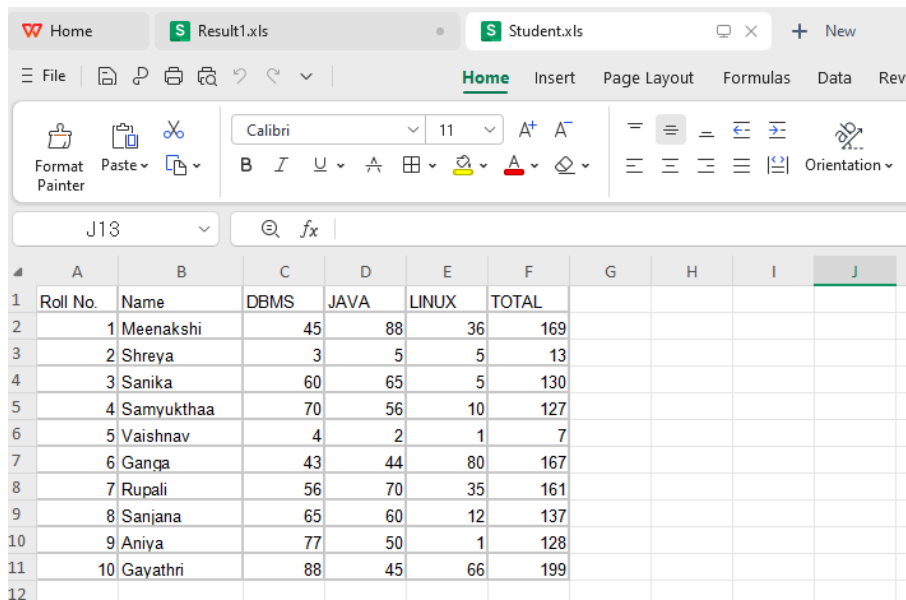
PRACTICAL 6

Write and test a program to select the number of students who have scored more than 60 in any one subject (or all subject)

Step 1:

Create 2 excel files Students.xls with 10 records and Result.xls empty

If extension of file is .Xlsx (check in properties) then save as change the type as



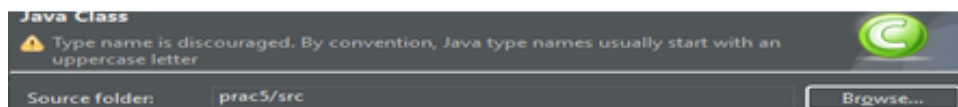
The screenshot shows an Excel spreadsheet with the following data:

Roll No.	Name	DBMS	JAVA	LINUX	TOTAL
1	Meenakshi	45	88	36	169
2	Shreya	3	5	5	13
3	Sanika	60	65	5	130
4	Samyukthaa	70	56	10	127
5	Vaishnav	4	2	1	7
6	Ganga	43	44	80	167
7	Rupali	56	70	35	161
8	Sanjana	65	60	12	137
9	Aniya	77	50	1	128
10	Gayathri	88	45	66	199

This will change the extension to .xls

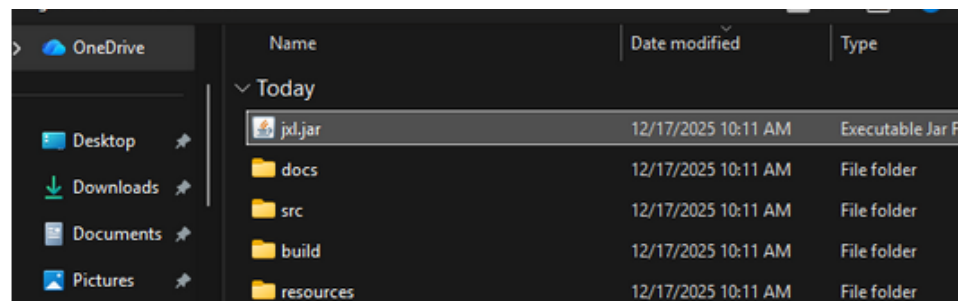
Step 2:

Create project



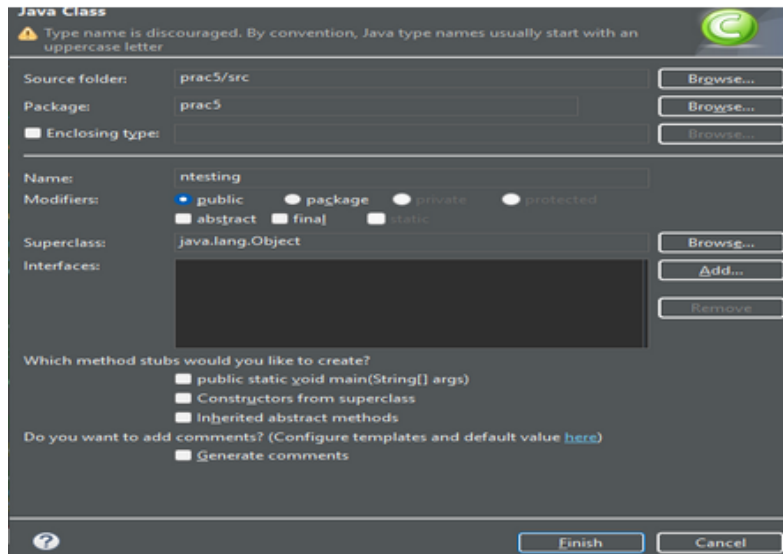
Step 3:

Download >> Add external jar



Step 4:

Create class>> NO Package(default package)>> No public static void main



Step 5:

Write code with all the libraries

Code :-

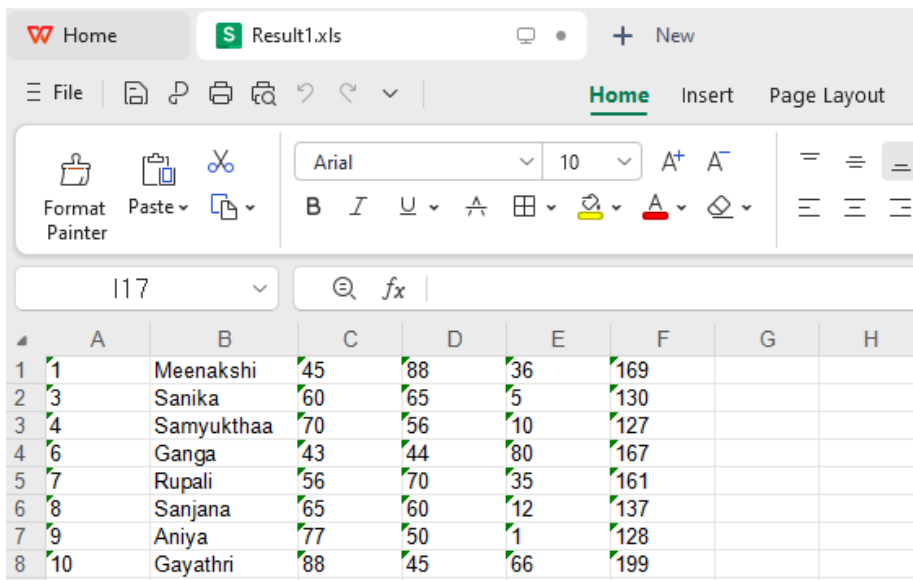
```
import jxl.write.Label;
import jxl.write.WritableSheet;
import jxl.write.WritableWorkbook;
import java.io.*;
import jxl.*;
import org.testng.annotations.*;
public class Student_Record {

    @Test
    public void testImportexport1() throws Exception {
        FileInputStream fi = new FileInputStream("C:\\Users\\student\\Desktop\\Student.xls");
        Workbook w = Workbook.getWorkbook (fi);
        Sheet s=w.getSheet(0);
        FileOutputStream fo = new FileOutputStream("C:\\Users\\student\\Desktop\\Result1.xls");
        WritableWorkbook ww = Workbook.createWorkbook (fo);
        WritableSheet ws = ww.createSheet ("result1", 0);
        int newRow = 0; // target row index
        for (int i = 1; i < s.getRows(); i++) // start from 1 (skip header)
        {
            int cMarks = Integer.parseInt(s.getCell(2, i).getContents()); // Column C
            int dMarks = Integer.parseInt(s.getCell(3, i).getContents()); // Column D
            int eMarks = Integer.parseInt(s.getCell(4, i).getContents()); // Column E
```

```
// If ANY subject has marks >= 60
if (cMarks >= 60 || dMarks >= 60 || eMarks >= 60)
{
    for (int j = 0; j < s.getColumns(); j++)
    {
        String value = s.getCell(j, i).getContents();
        Label l2 = new Label(j, newRow, value);
        ws.addCell(l2);
    }
    newRow++; // move to next output row
}
}
wwb.write();
wwb.close();
}}
```

Output:

Result1.xls



The screenshot shows a Microsoft Excel spreadsheet titled 'Result1.xls'. The spreadsheet contains a table with 8 rows and 7 columns (A-G). The data is as follows:

	A	B	C	D	E	F
1	1	Meenakshi	45	88	36	169
2	3	Sanika	60	65	5	130
3	4	Samyukthaa	70	56	10	127
4	6	Ganga	43	44	80	167
5	7	Rupali	56	70	35	161
6	8	Sanjana	65	60	12	137
7	9	Aniya	77	50	1	128
8	10	Gayathri	88	45	66	199

```
[RemoteTestNG] detected TestNG version 7.11.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
PASSED: Student_Record.testImportexport1

=====
Default test
Tests run: 1, Failures: 0, Skips: 0
=====

=====
Default suite
Total tests run: 1, Passes: 1, Failures: 0, Skips: 0
=====
```